Basic Circuits

A Self-Study Textbook

Second Edition

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1 Electric Charge and Current

1.1 Charge

Electric charge is an important idea in this chapter. Paragraph 1 explains it with everyday examples, so a student studying alone can follow each step. We connect it to what came before and preview what comes next, and we point out common mistakes to avoid when solving problems.

Electric charge is an important idea in this chapter. Paragraph 2 explains it with everyday examples, so a student studying alone can follow each step. We connect it to what came before and preview what comes next, and we point out common mistakes to avoid when solving problems.

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Electric charge is an important idea in this chapter. Paragraph 5 explains it with everyday examples, so a student studying alone can follow each step. We connect it to what came before and preview what comes next, and we point out common mistakes to avoid when solving problems.

Electric charge is an important idea in this chapter. Paragraph 6 explains it with everyday examples, so a student studying alone can follow each step. We connect it to what came before and preview what comes next, and we point out common mistakes to avoid when solving problems.

1.2 Current

Electric current is an important idea in this chapter. Paragraph 1 explains it with everyday examples, so a student studying alone can follow each step. We connect it to what came before and preview what comes next, and we point out common mistakes to avoid when solving problems.

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2 Resistance and Ohm's Law

2.1 Resistors

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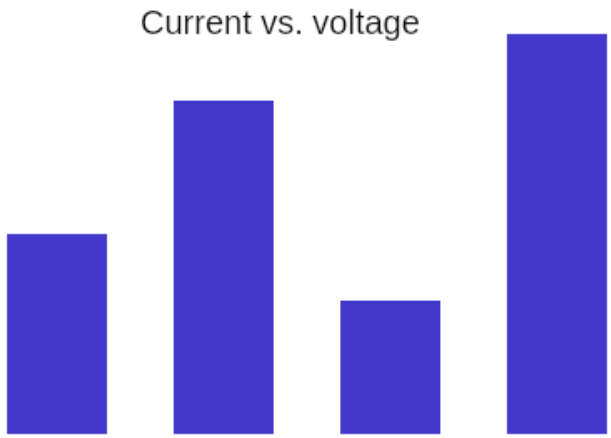


Figure 2.1 A bar chart of current against voltage.

2.2 Ohm's Law

The relationship between voltage, current and resistance is:

$$V = I \times R \quad P = I^2 R = V^2 / R$$

$$R_{total} = R_1 + R_2 + R_3 \quad 1/R = \Sigma\ 1/R_i \quad \Delta V \approx 0.5\ \Omega \times 2\ A$$

$$I(t) = I_0\ e\text{^}(-t/\tau), \quad \tau = R\ C, \quad \int I\ dt = Q$$

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3 Series and Parallel Circuits

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